

PathfinderAI: An Intelligent Comparative Analysis System for Pathfinding Algorithms with Multi-Method Evaluation

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Abstract

Abstract — Pathfinding algorithms are widely used in domains where both route quality and computational cost matter (e.g., autonomous navigation and game AI). PathfinderAI was developed to support *optimal evaluation* of pathfinding outcomes by combining interactive visualization with multi-factor, multi-method result analysis. Implemented in the Unity Engine, the system executes four pathfinding algorithms across six procedurally generated, solvable map types (Balanced, Cost-Heavy, Maze, Open Field, Heuristic Trap, and Random Chaos) and visualizes each algorithm’s step-by-step expansion on a 15×15 grid, including animated agent traversal of the discovered path. The platform captures core performance metrics—path cost, step count, visited nodes, and memory usage—and compares algorithm outputs using both conventional approaches (IF-statement decision logic and data-structure-based ranking lists) and AI-assisted approaches. AI integration includes local Small Language Models via koboldcpp with GGUF model support and fallback handling, alongside cloud Large Language Model benchmarking through external commercial APIs (including OpenAI GPT-4). Benchmarks were conducted on an RTX 2060 (6GB) system with 16GB RAM. By unifying visualization, metric capture, and multiple evaluation paradigms in one environment, PathfinderAI serves as a practical educational tool and a structured research testbed for systematic algorithm and SLM benchmarking, with future extensions targeting ML-enhanced SLMs that can directly propose optimal paths.